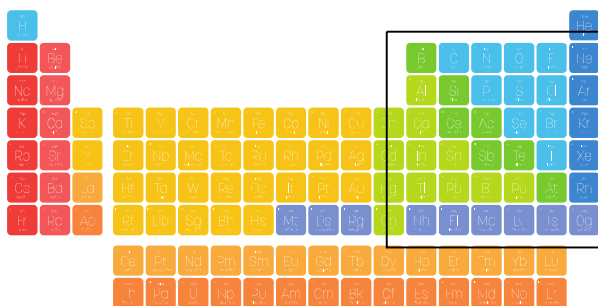




JEE (MAIN+ADVANCED)

Organic Chemistry

ENTHUSIAST COURSE



STUDY MATERIAL

Polymers

English Medium

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04**Polymers**

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IMPORTANT NOTE

Polymers

INTRODUCTION TO POLYMERS; DEFINITION OF POLYMERS; CLASSIFICATIONS OF POLYMERS; SOURCE OF

The term polymer is used to describe a very large molecule that is made up of many repeating small molecular units. These small molecular units from which the polymer is formed are called monomers. The chemical reaction that joins the monomers together is called polymerisation. Starting from n molecules of a compound M , linking in a linear manner will form polymer $x-M-(M)_{n-2}-M-y$. The nature of linkages at the terminal units i.e. $M-x$ and $M-y$ depends upon the mode of reaction used in making the polymers.

1.0 HOMOPOLYMERS AND COPOLYMERS

Polymers which are formed by only one type of monomer are called Homopolymers. Some examples of homopolymers and their monomers are given below:

Homopolymer	Monomer
Starch	Glucose
Cellulose	Glucose
Glycogen	Glucose
Dextrin	Glucose
Inulin	Fructose
Polyethylene	Ethylene
Polyvinyl chloride	Vinyl chloride
Teflon	Tetrafluoro ethylene
Nylon-6	Caprolactam
Polystyrene	Styrene
Orlon (Acrilan)	Acrylonitrile
Plexiglas (Lucite)	Methyl methacrylate
Polyvinyl acetate	Vinyl acetate

Polymers, which are formed by more than one type of monomers are known as copolymers. Some examples are given below in the table:

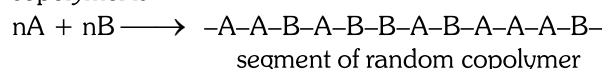
Copolymer	Monomers
Saran	Vinyl chloride and vinylidene chloride
SAN	Styrene and acrylonitrile
ABS	Acrylonitrile, butadiene and styrene
Butyl rubber	Isobutylene and Isoprene
Buna-S, SBR	Styrene and Butadiene
Buna-N, NBR	Acrylonitrile and Butadiene
Nylon-66	Hexamethylenediamine and Adipic acid
Terylene	Terephthalic acid and ethylene glycol

2.0 TYPES OF COPOLYMERS

Depending upon the distribution of monomer units, the following types of copolymers are possible.

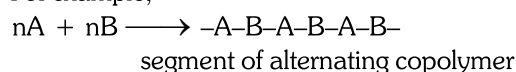
(1) Random Copolymer

If the monomer units have random distribution throughout the chain, it is called random copolymer. For example, if the monomer A and monomer B undergo copolymerisation then the structure of the random copolymer is



(2) Alternating Copolymer

If the two monomer units occur alternatively throughout the polymer chain, it is said to be alternating copolymer. For example,

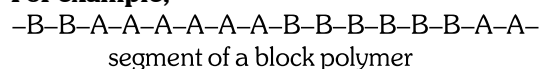


The exact distribution depends upon the proportion of the two reactant monomers and their relative reactivities. In practice neither perfectly random nor perfectly alternating copolymers are usually formed. However, most copolymers tend more towards alternating type but have many random imperfections.

(3) Block copolymer

Polymers in which different blocks of identical monomer units alternate with each other are called block copolymers.

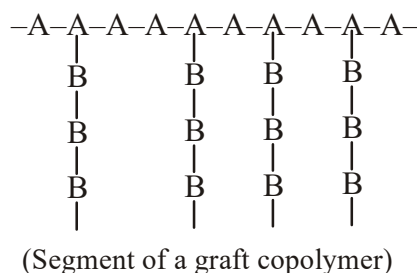
For example,



Block copolymer can be prepared by initiating the radical polymerisation of one monomer to grow homopolymer chains, followed by addition of an excess of the second monomer.

(4) Graft copolymer

Polymers in which homopolymer branches of one monomer unit are grafted onto a homopolymer chain of another monomer unit are called graft co-polymers. For example:



Graft copolymers are prepared by γ -irradiation of a homopolymer chain in the presence of a second monomer. the high energy radiation knock out H-atoms from the homopolymer chain at random points thus generating radical sites that can initiate polymerisation of the second monomer.

3.0 CLASSIFICATION OF POLYMERS; MOLECULAR FORCES; DEGRADATIONS; MODE OF POLYMERISATION

Polymers are classified in following ways :

(I) CLASSIFICATION BASED UPON SOURCE

(1) Natural polymers

Polymers which are obtained from animals and plants are known as natural polymers. Examples of natural polymers are given below.

Natural polymer

Monomers

- | | |
|--------------------------------------|-----------------------------------|
| 1. Polysaccharide | Monosaccharide |
| 2. Proteins | α -L-Amino acids |
| 3. Nucleic acid | Nucleotide |
| 4. Silk | Amino acids |
| 5. Natural Rubber (cis polyisoprene) | Isoprene (2-Methyl-1,3-butadiene) |
| 6. Gutta purcha (trans polyisoprene) | Isoprene |

Natural polymers which take part in metabolic processes are known as biopolymers. Examples are polysaccharides, proteins, RNA and DNA.

(2) Semisynthetic polymers

Polymers which are prepared from natural polymers are known as semisynthetic polymers. Most of the semisynthetic polymers are prepared from cellulose. Examples are: cellulose acetate, cellulose nitrate, cellulose xanthate and Rayon.

(3) Synthetic polymers

Man-made polymers, i.e. polymers prepared in laboratory are known as synthetic polymers. Example are : PVC, polyethylene, polystyrene, nylon-6, nylon-66, nylon-610, terylene, synthetic rubbers etc.

(II) CLASSIFICATION BASED UPON SHAPE**(1) Linear polymers**

Polymer whose structure is linear is known as linear polymer. The various linear polymeric chains are stacked over one another to give a well packed structure.



The chains are highly ordered with respect to one another. The structure is close packed in nature, due to which they have high densities, high melting point and high tensile (pulling) strength. Linear polymers can be converted into fibres.

Note :

- (i) All fibers are linear polymers. Examples are cellulose, silk, nylon, terylene etc.
- (ii) Linear polymers may be condensation as well as addition polymers. Examples are cellulose, polypeptide, nucleic acid, nylon, terylene etc.

(2) Branched chain polymers

Branched chain polymers are those in which the monomeric units constitute a branched chain. Due to the presence of branches, these polymers do not pack well. As a result branched chain polymers have lower melting points, low densities and tensile strength as compared to linear polymers.

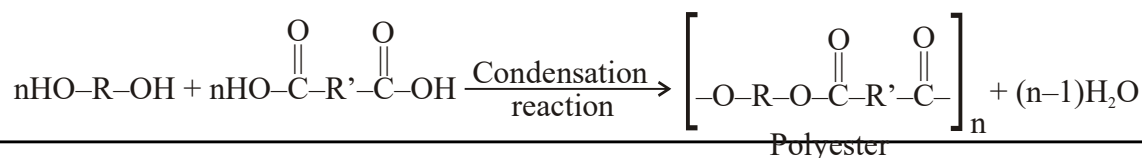
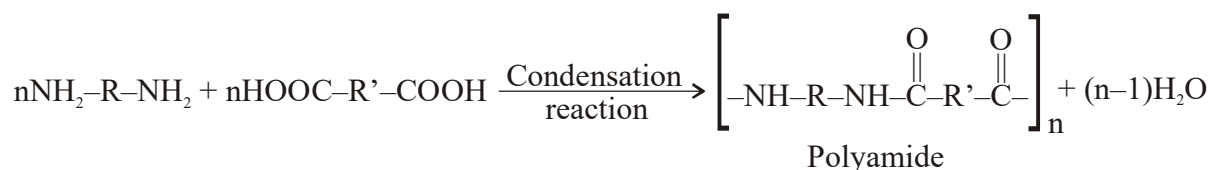
Branched chain polymers may be formed due to addition as well as condensation polymerisation. Examples are amylopectin, glycogen, low density polyethylene and all vulcanised rubbers.

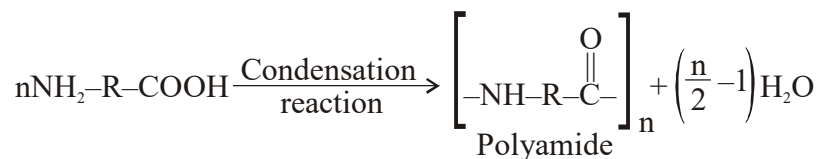
**(3) Cross-linked or Three Dimensional network polymers**

In these polymers the initially formed linear polymeric chains are joined together to form a three dimensional network structure. These polymers are hard, rigid and brittle. Cross-linked polymers are always condensation polymers. Resins are cross linked polymers.

**(III) CLASSIFICATION BASED UPON SYNTHESIS****(1) Condensation polymerisation**

- (i) They are formed due to condensation reactions.
- (ii) Condensation polymerisation is also known as step growth polymerisation.
- (iii) For condensation polymerisation, monomers should have at least two functional groups. Both functional groups may be same or different.
- (iii) Monomers having only two functional group always give linear polymer.
For example,





- (iv) Condensation polymers do not contain all atoms initially present in the monomers. Some atoms are lost in the form of small molecules.
- (v) Monomer having three functional groups always gives cross-linked polymer.

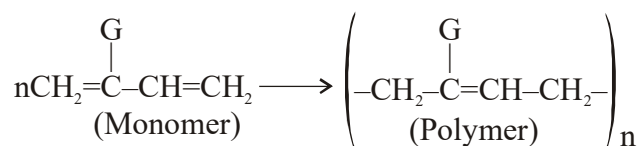
Examples are : Urea-formaldehyde resin, phenol-formaldehyde resin.

(3) Addition polymerisation

- (i) Polymers which are formed by addition reaction are known as addition polymers.
- (ii) If monomer is ethylene or its derivative, then addition polymer is either linear polymer or branch-chain polymer. **Examples are :** polystyrene, polytetrafluoroethylene, polyacrylonitrile etc.

- (iii) If monomer is 1, 3-butadiene or 2-substituted-1, 3-butadiene $\left(\text{CH}_2 = \underset{\text{G}}{\text{C}} - \text{CH} = \text{CH}_2 \right)$, then

polymer is always branched chain polymer.



(Monomer)

(i) G = H: 1,3 - Butadiene

(ii) G = CH₃ : 2 - Methyl-1,3-butadiene or isoprene

(iii) G = Cl ; 2 -Chloro-1,3-butadiene or chloroprene

- (iv) Addition polymers retain all the atoms of the monomer units in the polymer.

- (v) Addition polymerisation takes place in three steps :

Initiation, chain propagation and chain termination.

- (vi) Addition polymers are called as chain growth polymers.

(Polymer)

(i) Polybutadiene

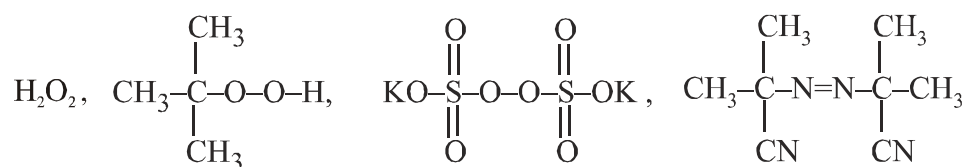
(ii) Polyisoprene

(ii) Polychloroprene (Neoprene)

4.0 MECHANISM OF FREE RADICAL POLYMERISATION QUESTIONS

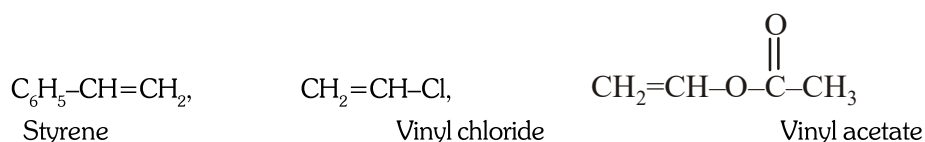
(A) Radical Polymerisation :

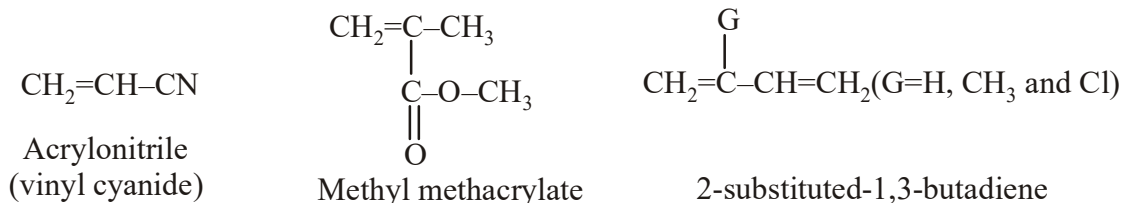
- (i) Radical polymerisation takes place in the presence of radical initiators. The radical initiator may be any of the following :



- (ii) Reaction intermediate of radical polymerisation is a free radical.

- (iii) Radical polymerization has more chance for those monomers whose free radicals are more stable. Examples are :

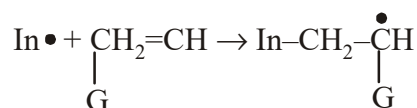
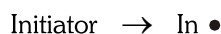




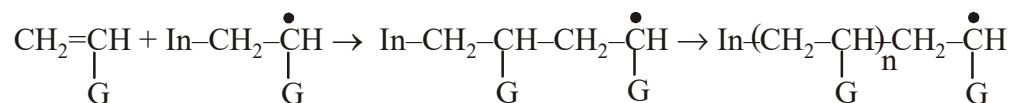
- (iv) Radical polymer has linear as well as branched chain structure
Most of the commercial addition polymers are vinyl polymers obtained from alkenes and their derivatives $\text{CH}_2=\text{CH}$ [G is H, C_6H_5 , R etc.]
- $$\begin{array}{c} \text{CH}_2=\text{CH} \\ | \\ \text{G} \end{array}$$

This type of polymerisation is performed by heating the monomer with only a very small amount of the initiator or by exposing the monomer to light. the general mode of radical polymerisation of vinyl monomers is depicted below :

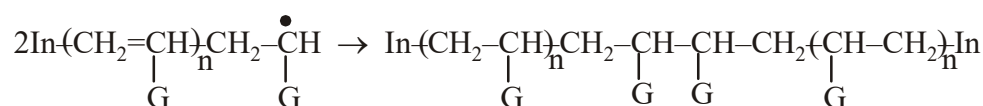
Chain initiation step :



Chain propagating step :



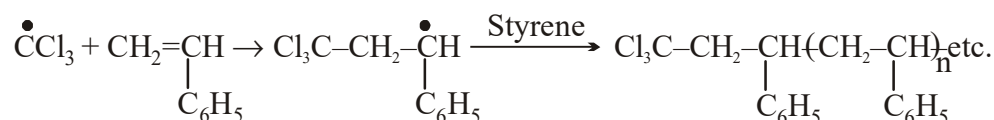
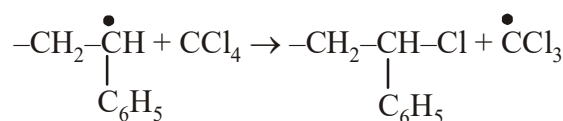
Chain terminating step :



Polymer

In vinylic polymerisation, various other reaction of free radicals with some other compounds present may compete with the parent addition chain reactions. One such reaction takes place with molecules that can react with the growing chain to interrupt the further growth. This leads to the lowering of the average molecular mass of the polymer. Such reagents are called as chain transfer agents and include CCl_4 , CBr_4 etc.

For example, in the presence of CCl_4 , styrene polymerises to form polystyrene of a lower average molecular mass which also contains some chlorine what happens here is that growing polystyrene radical which normally would add on a monomer reacts with the chain transfer agent to end the original chain and produces a new radical. The latter initiates a new polymerisation chain and thereby forms a new polymer as depicted below.

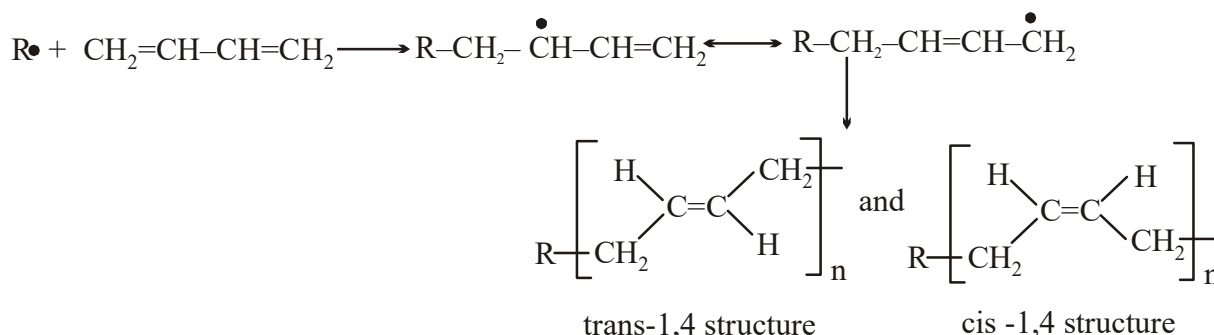


If the chain transfer agent a radical, which is highly unreactive, the reaction chain gets terminated such a compound thus inhibits or arrests polymerisation. Many amines, phenols, quinones etc. act as inhibitors. So, even traces of certain impurities, which can act as chain transfer agent or an inhibitor can interfere with the original polymerisation chain reaction. Hence, the monomers could be free from such inhibitors.

In case the alkene is a diene, the following kinds of polymerisation is possible :

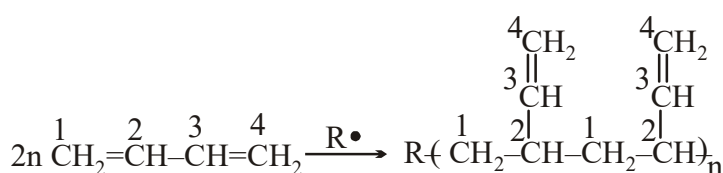
(1) **1,4 - polymerisation.**

When the polymerisation takes place at C_1 and C_4 of butadiene, an unbranched polymer is formed. This product is different from that formed from an alkene having a double bond, which at each of its carbons is substituted by different groups and hence can exist either as trans-polybutadiene or cis-polybutadiene or a mixture as shown below.



(2) **1,2-Polymerisation**

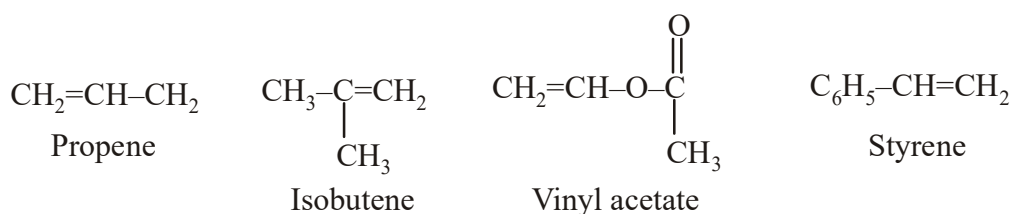
Alternatively, 1,3 -butadiene can undergo polymerisation at C_1 and C_2 to yield the polymeric product, polyvinyl polythene.



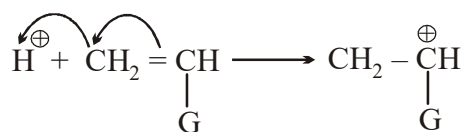
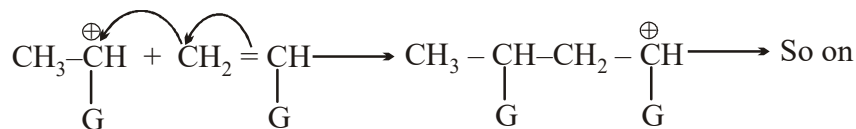
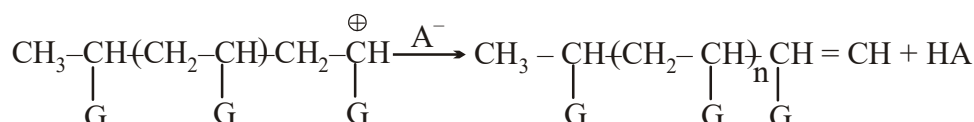
The double bonds in these initial polymers can be linked by further treatment with chemicals to modify the properties of the polymers. These reactions form the basis of the formation of rubber.

(B) **Cationic Polymerisation :**

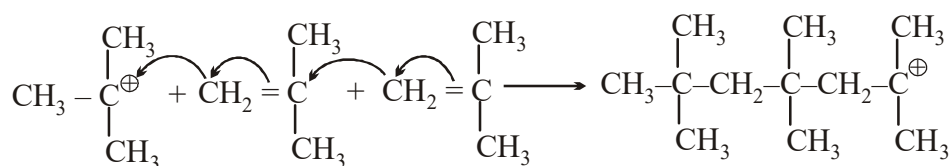
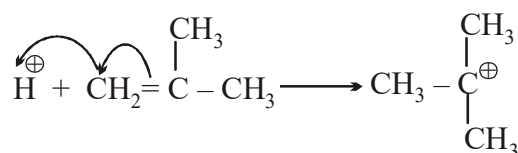
- (i) Polymerisation which is initiated by an electrophile is known as cationic polymerisation.
- (ii) Reaction intermediate of cationic polymerisation is a carbocation.
- (iii) Carbocations can undergo rearrangement leading to the formation of a more stable carbocation.
- (iv) The electrophile commonly used for initiation is $BF_3 \cdot OEt_2$.
- (v) Monomers that are best able to undergo polymerisation by a cationic mechanism are those with electron - donating substituents that can stabilise the carbonocation. Some examples are:



- (v) It is terminated by a base.
- Thus, when the initiator is cationic in nature, it would generate a cationic intermediate on addition to the double bond for propagating the addition chain process and is termed as cationic addition polymerisation. The process is initiated by an acid. The stages of polymerisation are depicted below.

Chain initiation step :**Chain propagating step :****Chain terminating step :**

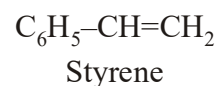
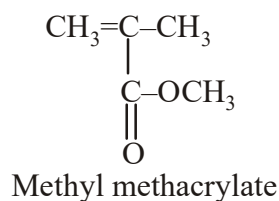
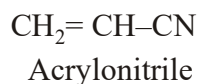
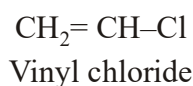
Cationic polymerisation is facilitated in monomers containing electron-releasing groups. Thus, isobutylene undergoes cationic polymerisation easily as it has two electron-releasing $-\text{CH}_3$ groups that will stabilize the intermediate carbo cation.



(Butyl Rubber)

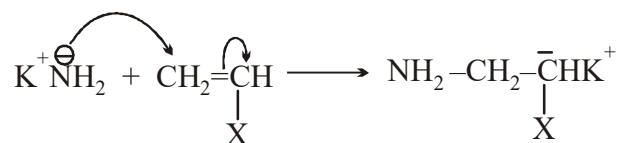
(C) Anionic Polymerisation :

- (i) Anionic polymerisation takes place in the presence of base or nucleophile, which is initiator in this polymerization.
- (ii) Reaction intermediate in propagation steps are carboanion.
- (iii) The suitable initiator can be NaNH_2 or RLi .
- (iv) Those monomers undergo anionic polymerisation reaction whose anion is stable.
Example of monomers are :

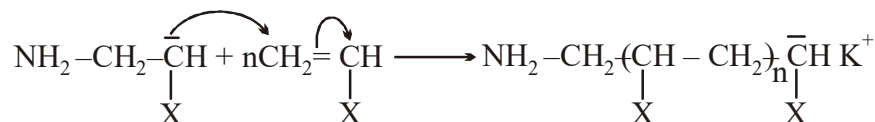


- (v) Anionic polymerisation always gives linear polymer.
- (vi) Anionic polymerisation terminated by an acid.
The formation of polystyrene from styrene in the presence of potassium amide is an important example of this category of polymerisation. The mode of anionic polymerisation is depicted below:

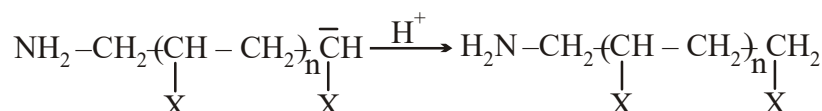
Chain initiation step :



Chain propagating step :



Chain terminating step :



(D) Ziegler- Natta polymerisation :

- (i) Addition polymerisation which takes place in the presence of Ziegler- Natta catalyst $[(\text{C}_2\text{H}_5)_3\text{Al}$ and $\text{TiCl}_4]$ is known as Ziegler- Natta polymerisation or coordination polymerisation.
- (ii) Ziegler- Natta polymerisation always gives linear, stereo-regular polymers.
- (iii) Ziegler- Natta catalyst revolutionised the field of polymer chemistry because they allow the synthesis of stronger and stiffer polymers (due to linear geometry) that have greater resistance to cracking and heat. High density polyethylene is prepared using a Ziegler- Natta catalyst.

5.0 CLASSIFICATION BASED ON INTERMOLECULAR FORCES (SECONDARY FORCES)

- (i) Intermolecular forces present between polymeric chains are (a) Van der waals forces (b) Hydrogen bonds and (c) Dipole - dipole attractions.
- (ii) Mechanical properties such as tensile strength, elasticity, toughness etc. depend upon the secondary forces present between the polymeric chains.
- (iii) Magnitude of secondary forces depends upon the size of the molecule and the number of functional groups along the polymeric chains.

Magnitude of secondary forces is directly proportional to the length of the polymeric chain. On the basis of magnitude of secondary forces, polymers can be divided into the following following categories.

(1) Elastomes

An elastomer is a plastic that stretches and then reverts back to its original shape. It is randomly oriented amorphous polymer. It must have some cross-links so that the chains do not slip over one another. Very weak Van der waal forces are present in between polymeric chains.

When elastomers are stretched, the random chains stretch out, but there are insufficient Van der Waal forces to maintain them in that configuration and position. When the stretching force is removed, they go back to their random shape. Elastomers have the ability to stretch out over ten times their normal length. Important examples are vulcanized rubbers.

Note : Addition polymers obtained from butadiene and its derivatives are elastomers.

(2) Fibres

Fibres are linear polymers in which the individual chains of a polymer are held together by hydrogen bonds and / or dipole-dipole attraction. In the fibres, the polymeric chains are highly ordered with respect to one another.

Due to strong intermolecular forces of attraction and highly ordered geometry, fibres have high tensile strength and least elasticity. they have crystalline character and have high melting points and low solubility. Examples are cellulose, nylon, terylene, wool, silk etc.

Note :

- (i) Condensation polymers formed from bifunctional monomers are fibres in character.
- (ii) Addition polymers of alkene derivatives having strong- I group are fibres in character.

(3) Thermoplastic Polymers

Thermoplastic polymers are polymers that have both ordered crystalline regions (the regions of the polymer in which the chains are highly ordered with respect to one another) and amorphous, non crystalline regions (the regions of the polymer in which the chains are randomly oriented).

The intermolecular forces of attraction are in between elastomers and fibres. There are no cross links between the polymeric chains. Thermoplastic polymers are hard at room temperature, but when they are heated, the individual chains can slip past one another and the polymer become soft and viscus. this soft and viscous material become rigid on cooling. The process of heating softening and cooling can be repeated as many times as desired without any change in chemical composition and mechanical properties of the plastic. As a result, these plastics can be moulded into toys, buckets, telephone and television cases. Some common examples are : polyethene polypropylene, polystyrene, polyvinyl chloride, teflon etc.

Note : Addition polymers obtained from ethylene and ethylene derivatives are thermoplastic polymers.

(4) Thermosetting Polymers

Polymers which become hard on heating are called thermosetting polymers. thermosetting polymers can be heated only once when it permanently sets into a solid, which cannot be remelted by heating.

Thermosetting polymers are cross- linked polymers . Greater the degree of cross - linking that exist, the more rigid is the polymer. Cross-linking reduces the mobility of the polymer chains, causing them to be relatively brittle materials. the hardening on heating is due to the extensive cross-linking between different polymer chains to give a three dimensional network solid. Examples are : phenol formaldehyde resin, urea-formaldehyde resin ,melamine - formaldehyde resin.

6.0 DIFFERENCE BETWEEN THERMOSETTING POLYMERS AND THERMOPLASTIC

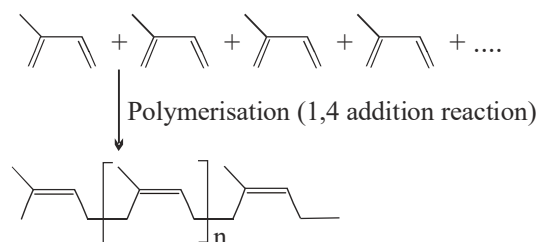
S.No.	Thermoplastic polymers	Thermosetting polymers
1.	Soften and melt on heating and become hard on cooling i.e. process is reversible	Become hard on heating and process is irreversible.
2.	Can be moulded and remoulded and reshaped.	They can be moulded once and cannot be remoulded or reshaped.
3.	They are addition polymers	They are condensation polymers.
4.	Structure is generally linear	Structure is cross - linked.

7.0 RUBBER

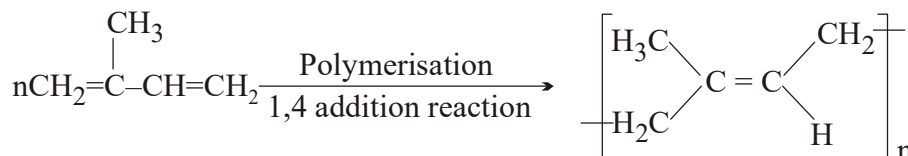
7.1 Natural Rubber

Natural rubber is obtained from nearly five hundred different plants but the main source is a braziliensis tree. It is obtained in the form of milky sap known as latex. This latex is coagulated with acetic acid and formic acid. The coagulated mass is then squeezed.

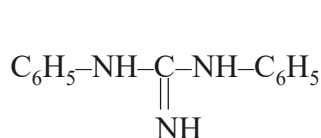
The raw natural rubber is a soft gummy and sticky mass. It is insoluble in water, dilute acids and alkalies but soluble in non-polar solvents. It has low elasticity and low tensile strength. Natural rubber is a polymer of 2-methyl-1,3-butadiene(isoprene). On average, a molecule of rubber contains 5000 isoprene units held together by head to tail. All the double bonds in rubber are cis, hence natural rubber is cis-polyisoprene.



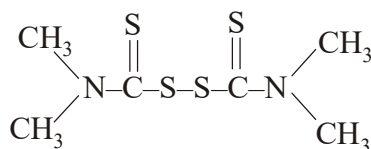
Gutta - percha is a naturally occurring isomer of rubber in which all the double bonds trans. Thus, gutta-percha is trans-polyisoprene.



It is harder and more brittle than rubber. It is the filling material that dentists use in root canal treatment. In order to give strength and elasticity to natural rubber, it is vulcanized. Heating of rubber with sulphur or sulphur containing compound at 150° C for few hours is known as vulcanisation. The essential feature of the vulcanisation is the formation of cross-linking between the polymeric chains. This cross-linking gives mechanical strength to the rubber. Vulcanisation process can be enhanced in the presence of certain organic compounds known as accelerators. The common accelerators are:



Diphenyl guanidine



Tetramethylthioureadisulph

In addition, fillers such as carbon black and zinc oxide are usually added to the crude rubber before vulcanisation in order to improve its wearing characteristics.

Natural rubber is used for making shoes, water - proof coats and golf balls. Vulcanised rubber is used for manufacture of rubber bands, gloves tubing and car tyres.

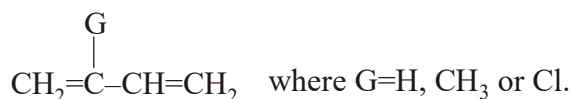
8.0 Examples Of Addition Polymerisation; Difference Between LDPE and HDPE; Examples Of Condensation Polymerisation

Polymers of 1,3 - butadienes are called synthetic rubbers because they have some of the properties of natural rubbers including the fact that they are water proof and elastic.

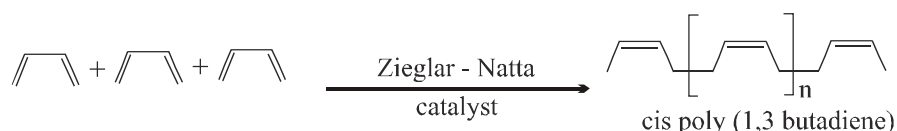
Synthetic rubbers have some improved properties. They are more flexible, tougher and more durable than natural rubber.

8.1 Homopolymers

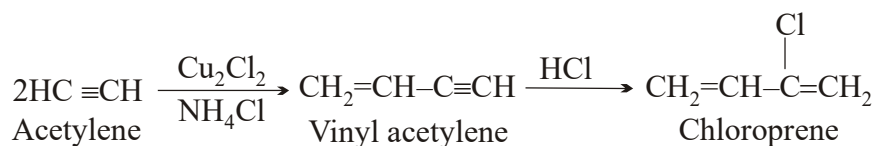
Monomer of this class is 2 - substituted - 1,3- butadienes.



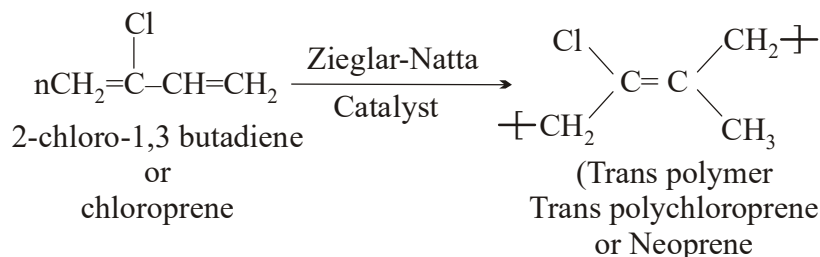
Polymerisation is always carried out in the presence of Ziegler-Natta catalyst which gives stereo regular polymers.



Neoprene was the first synthetic rubber manufactured on large scale. It is also called dieprene. Its monomer, chloroprene(2-chlorobutadiene) is prepared from acetylene.



Chloroprene undergoes free radical polymerisation to form neoprene (polychloroprene).



Many of the properties of neoprene are similar to natural rubber but neoprene is more resistant to action of oils, gasoline and other hydrocarbons. It is also resistant to sunlight, oxygen, ozone and heat. It is non-inflammable. It is used for making automobile and refrigerator parts, hoses for petrol and oil containers, insulation of electric wires and conveyor belts.

8.2 Copolymers

The following synthetic rubbers are example of copolymers.

Synthetic rubber

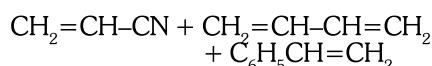
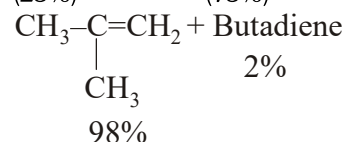
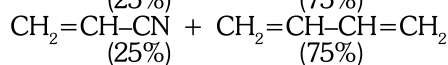
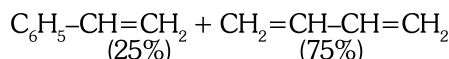
1. Buna-S, SBR (styrene-Butadiene rubber)

2. Buna-N, NBR (Nitrile-Butadiene rubber)

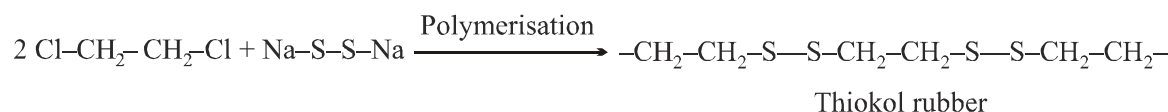
3. Butyl rubber

4. ABS; Acrylonitrile, Butadiene, Styrene

Monomers

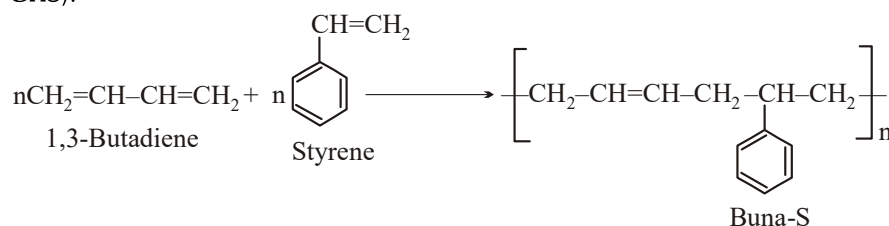


(a) **Thiokol** : Thiokol is made by polymerising ethylene chloride and sodium polysulphide.



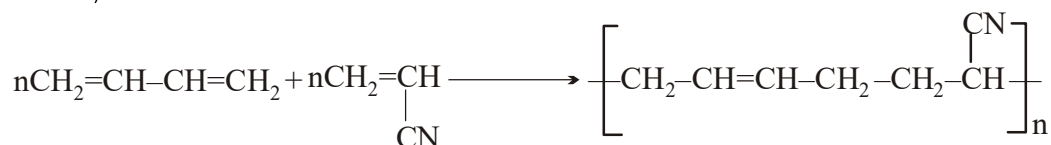
The repeating unit is $-\text{CH}_2-\text{S}-\text{S}-\text{CH}_2-$. Thiokol is chemically resistant polymer. It is used in the manufacture of hoses and tank linings, engine gaskets and rocket fuel when mixed with oxidising agents.

(b) **Buna-S (SBR : Styrene-butadiene rubber)** : Buna-S rubber is a copolymer of three moles of butadiene and one mole of styrene. In Buna-S, 'Bu' stands for butadiene, 'na' for symbol of sodium (Na) which is a polymerizing agent and 'S' stands for styrene. It is an elastomer (General purpose styrene Rubber or GRS).



Buna-S is generally compounded with carbon black and vulcanised with sulphur. It is extremely resistant to wear and tear and therefore used in the manufacture of tyres and other mechanical rubber goods. It is obtained as a result of free radical copolymerisation of its monomers.

- (c) **Buna-N** : It is obtained by copolymerisation of butadiene and acrylonitrile (General purpose Rubber acrylonitrile or GRA).



It is very rigid and is very resistant to action of petrol, lubricating oil and many organic solvents. It is mainly used for making fuel tanks.

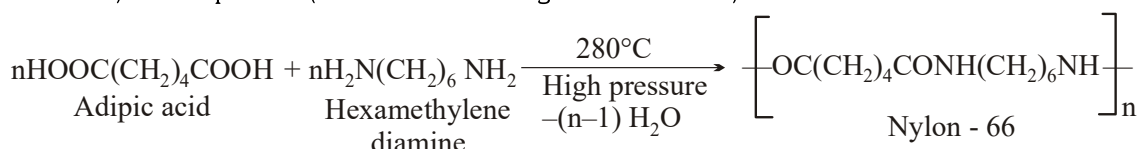
- (d) **Cold Rubber** : Cold rubber is obtained by polymerisation of butadiene and styrene at -18° to 5°C temperature in the presence of redox system. Cold rubber has a greater tensile strength and greater resistance of abrasion than SBR.

9.0 NYLON

Nylon is used as a general name for all synthetic fibres forming polyamides, i.e., having a protein like structure. A number is usually suffixed with the name 'nylon' which refers to the number of carbon atoms present in the monomers.

9.1 NYLON - 66 (Nylon six, six)

It is obtained by the condensation polymerisation of hexamethylenediamine (a diamine with six carbon atoms) and adipic acid (a dibasic acid having 6 carbon atoms).



9.2 NYLON-610 (Nylon six, ten)

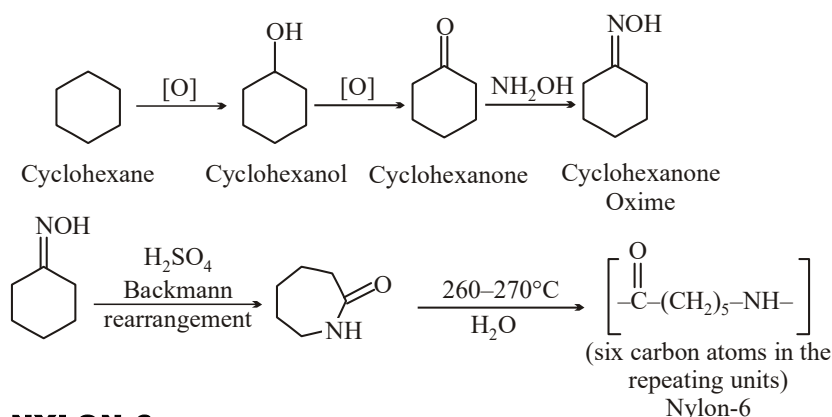
It is obtained by condensation polymerisation of hexamethylenediamine (six carbon atoms) and sebacic acid (a dibasic acid with 10 carbon atoms.)

Nylon fibres are stronger than natural fibres and so are used in making cords and ropes. The fibres are elastic, light, very strong and flexible. They have drip dry property and retain creases. It is inert towards chemicals and biological agents. It can be blended with wool. Nylon fibres are used in making garments, carpets, fabrics, tyre cords, ropes, etc.

9.3 NYLON-6 (Perlon L)

A polyamide closely related to nylon is known as perlon L (Germany) or Nylon - 6 (USA). It is prepared by prolonged heating of caprolactam at $260-270^\circ\text{C}$. It is formed by self condensation of a large number of molecules of amino caproic acid. Since, caprolactam is more easily available, it is used for polymerization, with is carried out in the presence of H_2O that first hydrolyses the lactam to amino acid. Subsequently, the amino acid can react with the lactam and the process goes on and onto form the polyamide polymer.

Caprolactam is obtained by Backmann rearrangement of cyclohexanone oxime.



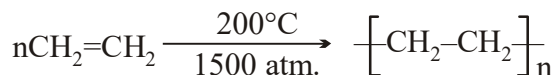
9.4 NYLON-2-NYLON-6

It is an alternating polyamide copolymer of glycine and amino caproic acid and is biodegradable.

10.0 POLYETHYLENE

Polyethylene is of two types :

- (a) **Low Density Poly Ethylene (LDPE)** : It is manufactured by heating ethylene at 200°C under a pressure of 1500 atmospheres and in the presence of traces of oxygen. This polymerisation is a free radical polymerisation.



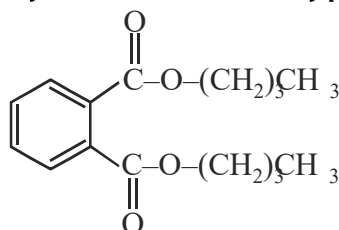
The polyethylene produced has a molecular mass of about 20,000 and has a branched structure. Due to this, polyethylene has a low density (0.92) and low melting point (110°C). That is why polyethylene prepared by free radical polymerisation is called low density polyethylene. It is a transparent polymer of moderate tensile strength and high toughness. It is widely used as a packing material and as insulation for electrical wires and cables.

- (b) **High Density Poly Ethylene (HDPE)** : It is prepared by the use of Ziegler - Natta catalyst at 160°C under pressure of 6 to 7 atmosphere.

The polymer is linear chain, hence it has high density (0.97) and has high melting point (130°C). That is why it is called high density polyethylene. It is a translucent polymer. It has greater toughness, hardness and tensile strength than low density polyethylene. It is used in the manufacture of containers (buckets, tubes), house wares, bottles and toys.

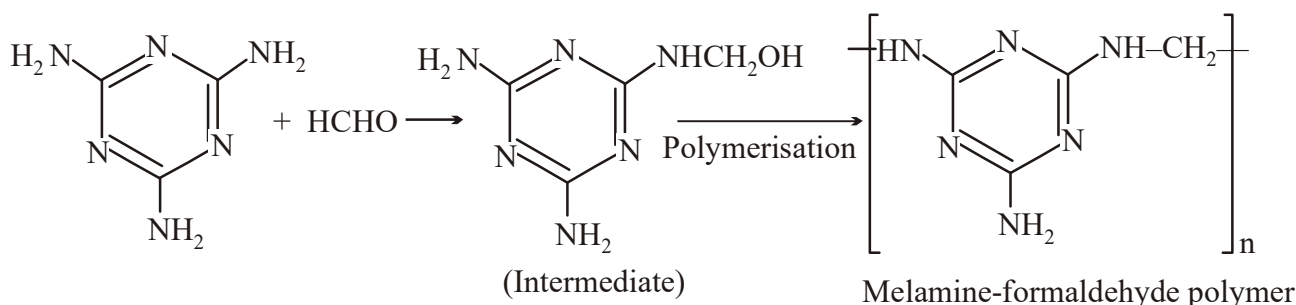
11.0 PLASTICISER

A plasticiser is an organic compound that dissolves in the polymer and allows the polymer chains to slide past one another. This makes polymer more flexible. Dibutylphthalate is a commonly used plasticiser.



12.0 MELAMINE - FORMALDEHYDE RESIN

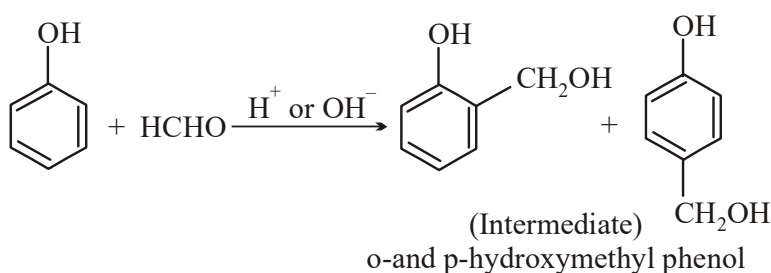
This resin is formed by condensation polymerisation of melamine and formaldehyde.

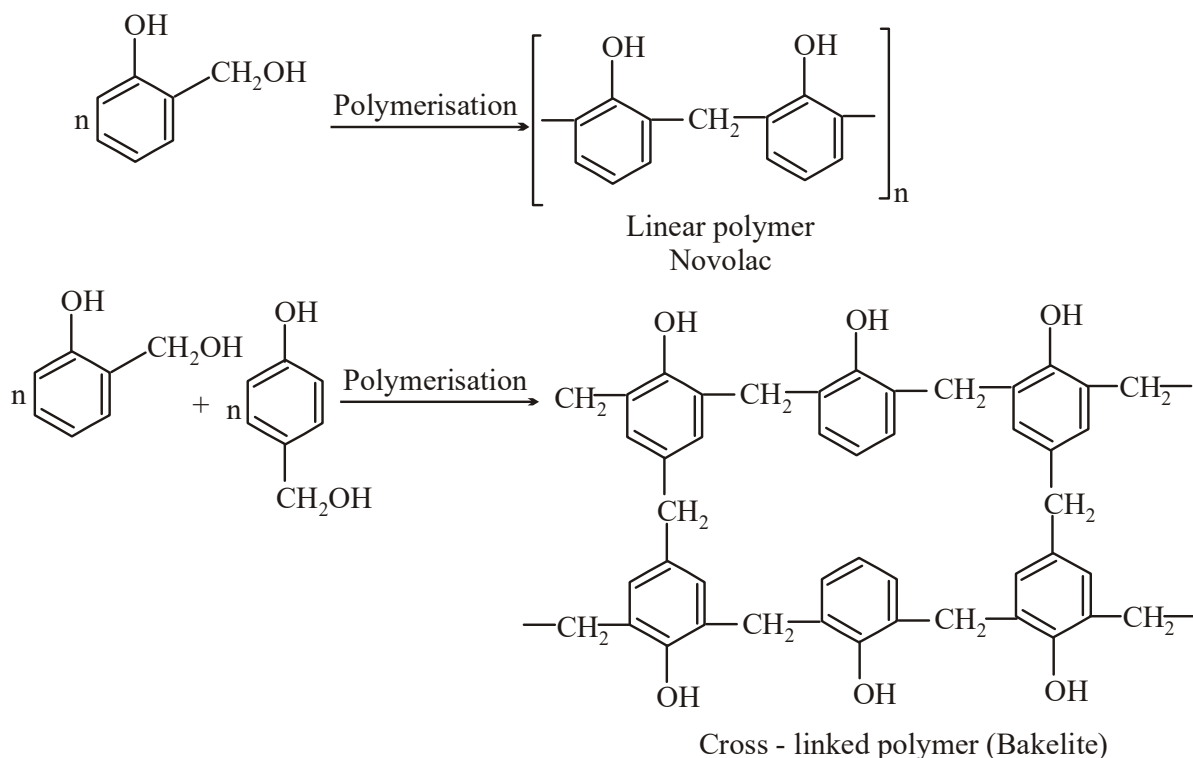


It is a quite hard polymer and is used widely for making plastic crockery under the name melamine. The articles made from this polymer do not break even when dropped from considerable height.

13.0 BAKELITE

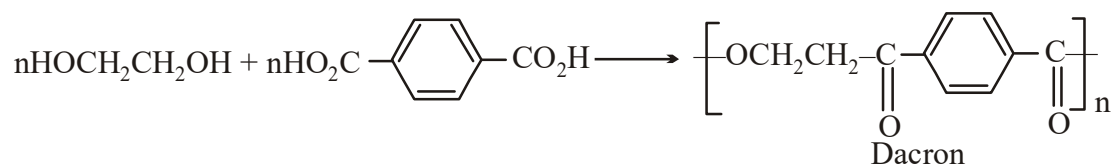
Phenol-formaldehyde resins are obtained by the reaction of phenol and formaldehyde in the presence of either an acid or a basic catalyst. The reaction starts with the initial formation of ortho and para-hydroxymethyl phenol derivatives, which further react with phenol to form compounds where rings are joined to each other with $-\text{CH}_2-$ groups. The reaction involves the formation of methylene bridges in ortho, para or both ortho and para positions. Linear or cross-linked materials are obtained depending on the conditions of the reaction.





14.0 POLYESTERS

Dacron is a common polyester, prepared using ethylene glycol and terephthalic acid. The reaction is carried out at 140° to 180° C in the presence of zinc acetate and Sb_2O_3 as catalyst.



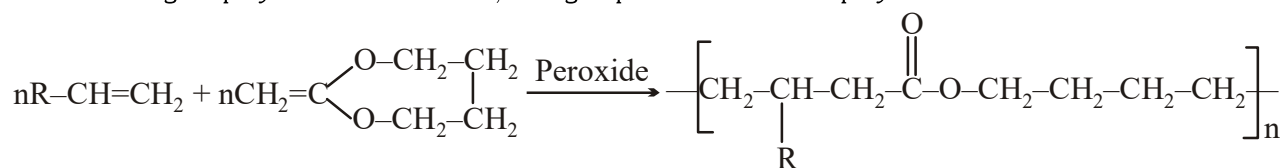
The terylene fibre (Dacron) is crease resistant and has low moisture absorption. It has high tensile strength. It is mainly used in making wash and wear garments, in blending with wool to provide better crease and wrinkle resistance.

15.0 BIODEGRADABLE POLYMERS

By far the largest use of synthetic polymers is as plastic. A major portion of it is used as throwaway containers and packing materials. Since plastics do not disintegrate by themselves, they are not biodegradable over a period of time. Non - biodegradability is due to the carbon-carbon bonds of addition polymers which are inert to enzyme catalysed reaction. These polymers create pollution problem.

Biodegradable polymers are the polymers that can be broken into small segments by enzyme catalysed reactions using enzymes produced by microorganisms. In biodegradable polymers, bonds that can be broken by the enzymes are inserted into the polymers. Therefore, when they are buried as waste, enzymes present in the ground can degrade the polymer.

One method involves inserting hydrolysable ester group into the polymer. For example, when acetal (I) is added during the polymerization of alkene, ester group is inserted into the polymeric chains.

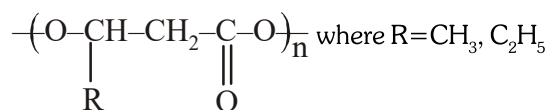
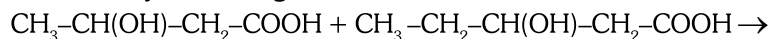


Ester linkage attacked by enzyme

Aliphatic polyesters are important class of biodegradable polymers. some examples are described below:

(1) Poly - Hydroxybutyrate-CO-β-Hydroxyvalerate (PHBV)

It is a copolymer of 3 - hydroxybutanoic acid and 3 hydroxypentanoic acid, in which the monomer units are connected by ester linkages.



The properties of PHBV vary according to the ratio of both the acids. 3-Hydroxybutanoic acid provides stiffness and 3-hydroxypentanoic acid imparts flexibility to the co-polymer. It is used in specialty packaging, orthopaedic devices and even in controlled drug release. When a drug is put in a capsule of PHBV, It is released only after the polymer is degraded. PHBV also undergoes bacterial degradation in the environment.

(2) POLY (GLYCOLIC ACID) AND POLY (LACTIC ACID)

They constitute commercially successful biodegradable polymers such as sutures. Dextron was the first bioadsorbable suture made for biodegradable polyesters for post - operative stitches.

16.0 MOLECULAR MASS OF POLYMER

Normally, a polymer contains chains of varying lengths and therefore, its molecular mass is always expressed as an average. In contrast, natural polymers such as protein contains chain of identical length and hence, have definite molecular mass.

The molecular mass of a polymer is expressed as

- (a) Number average molecular mass ($\overline{M}_n$)

$$\overline{M}_n = \frac{\sum N_i M_i}{\sum N_i}$$

where N_i is the number of molecules of molecular mass M_i

- (b) Weight average molecular mass ($\overline{M}_w$)

$$\overline{M}_w = \frac{\sum N_i M_i^2}{\sum N_i M_i}$$

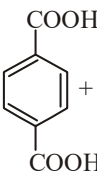
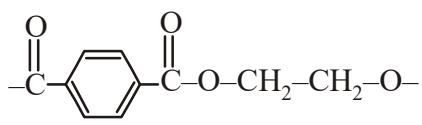
where N_i is the number of molecules of molecular mass M_i . Methods such as light scattering and ultracentrifuge depend on the mass of the individual molecules and yield weight average molecular masses. $\overline{M}_n$ is determined by employing methods which depend upon the number of molecules present in the polymer sample viz. colligative properties like osmotic pressure.

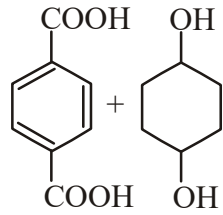
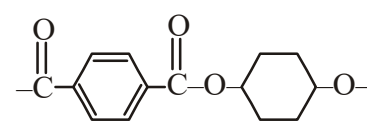
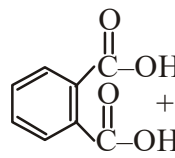
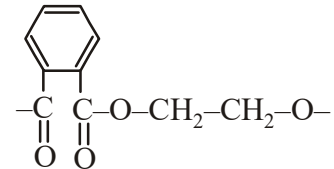
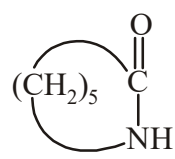
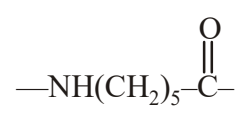
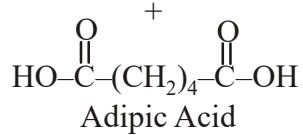
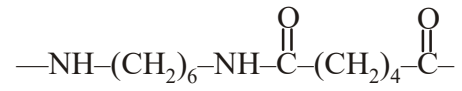
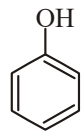
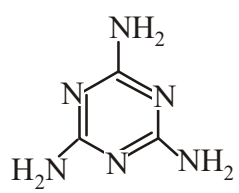
The ratio of the weight and number average molecular masses ($\overline{M}_w / \overline{M}_n$) is called Poly Dispersity Index (PDI). Some natural polymers, which are generally monodisperesed, the PDI is unity (i.e. $\overline{M}_w = \overline{M}_n$).

In synthetic polymers, which are always polydispersed, $\text{PDI} > 1$ because $\overline{M}_w$ is always higher than $\overline{M}_n$.

17.0 COMMON POLYMERS

	Monomer	Repeating unit	Polymer
1.	$\text{CH}_2=\text{CH}_2$ Ethylene	$\text{--CH}_2\text{--CH}_2\text{--}$	Polyethylene
2.	$\text{CH}_3\text{--CH}=\text{CH}_2$ Propene	$\begin{array}{c} \text{CH}_3 \\ \\ \text{--CH}_2\text{--CH--} \end{array}$	Polypropene
3.	$\text{C}_6\text{H}_5\text{--CH}=\text{CH}_2$ Styrene	$\begin{array}{c} \text{C}_6\text{H}_5 \\ \\ \text{--CH}_2\text{--CH--} \end{array}$	Polystyrene

4.	$\text{CF}_2=\text{CF}_2$ Tetrafluoroethylene	$-\text{CF}_2-\text{CF}_2-$	Polytetrafluoroethylene (PTFE), Teflon
5.	$\text{CH}_2=\text{CH}-\text{Cl}$ Vinyl chloride	$-\text{CH}_2-\underset{\text{Cl}}{\text{CH}}-$	Polyvinyl Chloride (PVC)
6.	$\text{CH}_2=\text{CH}-\text{CN}$ Vinyl cyanide or Acrylonitrile	$-\text{CH}_2-\underset{\text{CN}}{\text{CH}}-$	Polyvinyl cyanide, polyacrylonitrile, Orlon.
7.	$\text{CH}_2=\underset{\text{H}_3\text{C}}{\text{C}}-\overset{\text{O}}{\parallel}\text{C}-\text{O}-\text{CH}_3$ Methyl methacrylate	$-\text{CH}_2-\underset{\text{CH}_3}{\overset{\text{COOCH}_3}{\text{C}}}-$	Polymethyl methacrylate, Plexiglas, Lucite
8.	$\text{CH}_2=\text{CH}-\overset{\text{O}}{\parallel}\text{C}-\text{CH}_3$ Vinyl acetate	$-\text{CH}_2-\underset{\text{OCOCH}_3}{\text{CH}}-$	Polyvinyl Acetate
9.	$\text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ 1,3-butadiene	$-\text{CH}_2-\text{CH}=\text{CH}-\text{CH}_2-$	Polybutadiene, Buna rubber
10.	$\text{CH}_2=\text{CH}-\text{Cl}$ (vinyl chloride) + $\text{CH}_2=\text{CCl}_2$ (Vinylidene chloride)	$-\text{CH}_2-\underset{\text{Cl}}{\text{CH}}-\text{CCl}_2-\text{CH}_2-$	Saran
11.	$\text{C}_6\text{H}_5-\text{CH}=\text{CH}_2$ (Styrene) + $\text{CH}_2=\text{CH}-\text{CN}$ (acrylonitrile)	$-\text{CH}_2-\underset{\text{C}_6\text{H}_5}{\text{CH}}-\underset{\text{CN}}{\text{CH}}-\text{CH}_2-$	SAN
12.	$\text{CH}_2=\text{CH}-\text{CN} + \text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$ + $\text{C}_6\text{H}_5-\text{CH}=\text{CH}_2$	—	ABS
13.	$\text{CH}_2=\underset{\text{CH}_3}{\text{C}}-\text{CH}_2 + \text{CH}_2=\underset{\text{CH}_3}{\text{C}}-\text{CH}=\text{CH}_2$	—	Butyl rubber
14.	$\text{C}_6\text{H}_5-\text{CH}=\text{CH}_2 + \text{CH}_2=\text{CH}-\text{CH}=\text{CH}_2$	—	Buna -S, SBR
15.	$\text{CH}_2=\text{CH}-\text{CN} + \text{C}_6\text{H}_5-\text{CH}=\text{CH}_2$	—	Buna-N, NBR
16.	$\text{CH}_2=\underset{\text{Cl}}{\text{C}}-\text{CH}=\text{CH}_2$ Chloroprene	$-\text{CH}_2-\underset{\text{Cl}}{\text{C}}=\text{CH}-\text{CH}_2-$	Neoprene
17.	 + $\text{HO}-\text{CH}_2-\text{CH}_2-\text{OH}$		Poly(ethylene terephthalate, Terylene, Dacron or Mylar

18.  +  Kodel Polyester
19.  + HO-CH₂-CH₂-OH  Polyethylene
phthalate alkyd resin
(Glyptal)
20.  Caprolactam  Nylon-6
21. NH₂(CH₂)₆-NH₂ Hexamethylenediamine +  Adipic Acid  Nylon - 66
22.  + HCHO — Bakelite or resol
23. NH₂-C(=O)-NH₂ + HCHO — Urea-
formaldehyde resin
24.  + HCHO — Melamine Melamineresin

GOLDEN KEY POINTS

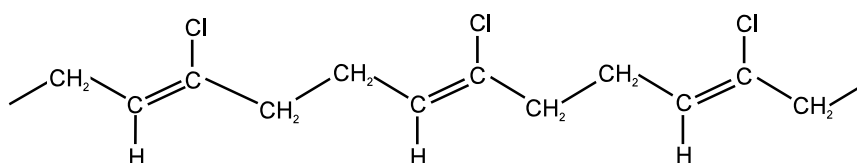
- Plasticizers fit between the polymer chains and thus weaken the attraction between the chains there by increasing the flexibility.
- Nylon-6(USA) or perlon (Germany) is prepared by prolonged heating of caprolactum at 540K.
- Saran is a copolymer of vinylidene chloride (85%) and vinyl chloride (15%).
- All surfactants consist of two characteristic groups, i.e., a polar head group which is water-soluble (hydrophilic group) and a non-polar hydrocarbon tail which is oil-soluble (hydrophobic or lipophilic group).

- Like soaps, detergents of the type, linear benzene sulphonate (LBS) in which the phenyl group is randomly attached to the various secondary positions of a long straight chain or n-alkyl hydrogen sulphates are 100% biodegradable.
- Congo red (azo dye) and martius yellow (nitro dye) are also called direct dyes.

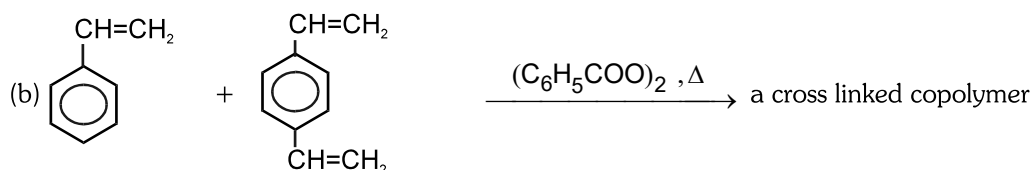
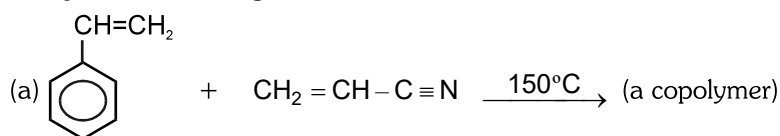
BEGINNER'S BOX-1

Complete the reaction :

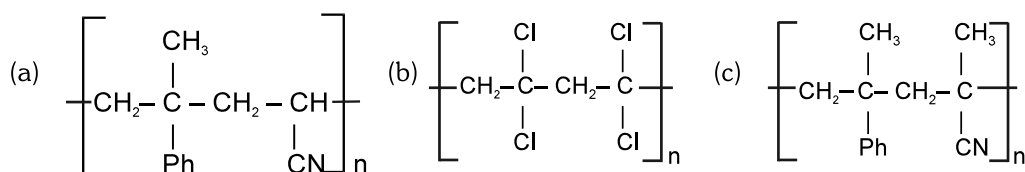
- $\text{CH} \equiv \text{CH} \xrightarrow[\Delta]{\text{NH}_4\text{Cl}, \text{Cu}_2\text{Cl}_2} \text{A} \xrightarrow{\text{HCl}} \text{B} \xrightarrow{\text{Polymerisation}} \text{C}$
- Give the structure of the reactant and product in the following reaction
 $\text{Natural rubber} \xrightarrow[\Delta]{\text{Sulphur}} \text{vulcanized rubber}$
- The partial structure of neoprene, a polymer is given below. Identify the monomer unit.



- Complete the following reactions

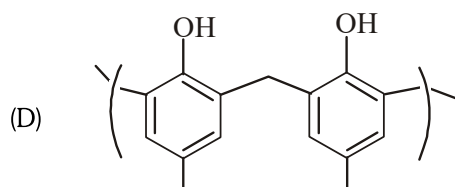
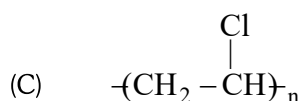
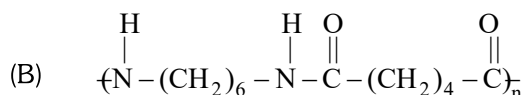
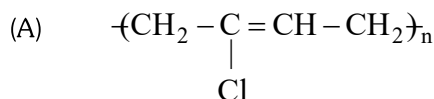


- Write the structures of monomers of the following polymers.



- Why nylon has very high melting point as compared to other polymers.
- Match the column.

Column I



Column II

(p) Thermoplastic polymer

(q) Thermosetting polymers

(r) Fibres

(s) Elastomer